Introduction

- This course introduces the fundamental concepts, methodologies, and applications of Machine Learning (ML) in today's data-driven computing environment.

- The course focuses on understanding and applying core aspects of machine learning, including types of learning paradigms, supervised and unsupervised learning algorithms, classification and regression techniques, model optimization, performance evaluation, ensemble methods, dimensionality reduction, and an introduction to deep learning and reinforcement learning.

Course Outcomes

- **CSE3201.1 ---->** Explain the fundamental concepts, history, need, applications, advantages, limitations, and challenges of Machine Learning, and identify appropriate machine learning paradigms for different problem domains.
- **CSE3201.2 ---->** Select and implement suitable machine learning algorithms and tools (classification, regression, clustering, dimensionality reduction) to solve real-world data-driven problems.
- **CSE3201.3 ---->** Apply supervised, unsupervised, and reinforcement learning techniques such as linear regression, logistic regression, Naïve Bayes, Decision Trees, SVM, KNN, clustering, and ensemble methods to build predictive models.
- **CSE3201.4 ---->** Analyze and compare feature extraction and dimensionality reduction techniques (PCA, LDA, ICA, SOM) and evaluate their impact on model performance and learning efficiency.
- **CSE3201.5 ---->** Evaluate and optimize machine learning models using performance metrics, validation techniques, regularization (L1/L2), optimization methods, and hyperparameter tuning to improve generalization and reduce overfitting.

Syllabus

- **Machine Learning**: Introduction and need; history; applications; advantages and disadvantages; limitations of machine learning; types of machine learning problems; classification and regression; supervised learning versus unsupervised learning; reinforcement learning;
- **Supervised Learning**: linear regression, logistic regression, cost function, activation and loss function, backpropagation, overfitting; underfitting and regularization (L1/L2), optimization techniques, hyperparameter tuning, performance metrics, validation methods, naïve Bayes, decision tree, SVM, KNN;
- **Ensemble Methods**: bagging, boosting, and stacking; Unsupervised Learning: K-means clustering, hierarchical clustering, DBSCAN, K-self organizing map;
- **Dimensionality Reduction**: linear discriminant analysis, principal component analysis (PCA), independent component analysis;
- **Deep Learning**: distributed representation, embedding, convolutional neural networks, recurrent neural networks, reinforcement learning.

Books

1. Stephen Marsland, Machine Learning – An Algorithmic Perspective, (2e), Chapman and Hall / CRC, 2014.

2. Tom Mitchell, Machine Learning, (1e), McGraw-Hill, 2017.

3. Christopher Bishop, Pattern Recognition and Machine Learning, (1e), Springer, 2009.

4. Ethem Alpaydin, Introduction to Machine Learning, The MIT Press, 2014, ISBN 978-0-262-02818-9.

5. Ian Good Fellow, Yoshua, Bengio, Aaron Courville, Deep Learning, MIT Press Book, 2016.

6. Mehryar Mohri, Afshin Rostamizadeh, Ameet Talwalkar, Foundations of Machine Learning, MIT Press (MA), 2012.

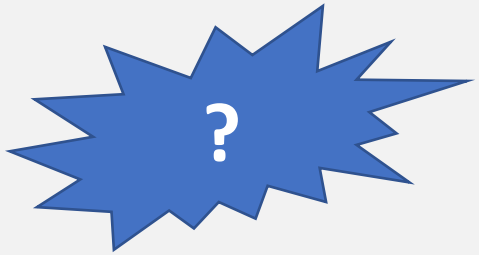
What is this object?

So, what happened here?

Human can learn from past experience and make decision of its own.

Let us ask the same question to him

What is this object?



But, he is a human being. So, He can observe and learn.

show him

CAR

CAR

BIKE

BIKE

Let us ask the same question again to him

What is this object?

CAR

CAR

BIKE

BIKE

CAR

What is Learning?

- Learning denotes changes in the system that are adaptive in the sense that they enable the system to do the same task or tasks drawn from the same population more effectively the next time. -- Simon, 1983
- Learning is making useful changes in our minds. -- Minsky, 1985
- Learning is constructing or modifying representations of what is being experienced. -- McCarthy, 1968
- Learning is improving automatically with experience. -- Mitchell, 1997
- Intelligence is based upon learning and learning is based upon past experience and knowledge about the problem.

What about a Machine?

Machines only follow instructions

We can ask a machine

- To perform an arithmetic operations such as
  - Addition
  - Multiplication
  - Division
  - Comparison
  - Print
  - Plotting a chart

It can not take decision of its own.

But, we can ask a machine to make a decision of its own.

What is Machine Learning?

We want a machine to act like a human.

To identify this object.

Recognize Face

Predict the price in future like 2030.

I ~~made~~ met him yesterday **Natural Language understand, and correct grammar.**

What is Machine Learning?

- What do we do?
- Just like, what we did to human like him.
- We need to provide experience to the machine.

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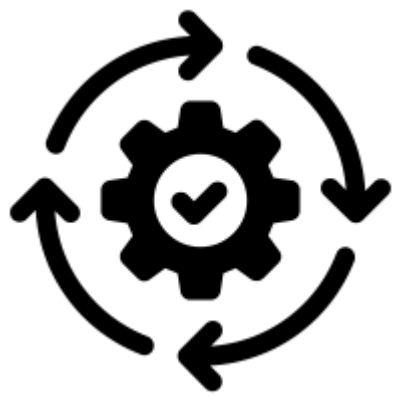

Devise algorithms and execute programs on the data with respect to the underlying target tasks.

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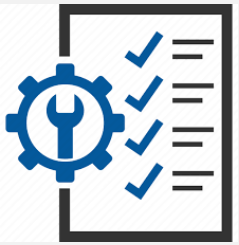
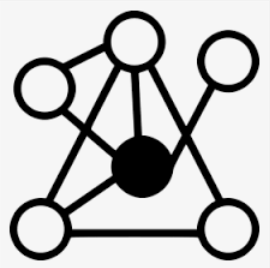
Identify required rules.



Extract required patterns

Identify relations.



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Machine can derive inferences from the data

Machine Learning

- Machine Learning (ML) is **a branch of artificial intelligence** that enables computers to **learn patterns from data and improve their performance automatically**, without being explicitly programmed for every task.
- In simple terms:
  - Instead of telling a computer exactly what to do, we give it data and let it figure out the rules on its own.
- A commonly used formal definition is:
  - "Machine Learning is the study of computer algorithms that improve automatically through experience (data)."
- Key ideas packed into this definition:
  - **Data-driven:** Learning happens from examples or historical data.
  - **Experience-based**: Performance improves as more data is processed.
  - **No explicit programming:** Rules are learned, not hard-coded

Machine Learning

- **Formal and standard definition of Machine Learning by Tom Mitchell**
  - "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E."

- **Task (T):** What the system is supposed to do (e.g., classify emails as spam or not spam)

- **Experience (E):** The data or past examples it learns from (e.g., previously labeled emails)

- **Performance Measure (P):** How we evaluate success (e.g., accuracy of classification)

- If accuracy improves as more emails are seen ⟶ Machine Learning is happening

Why is Machine Learning needed?

1. Traditional Programming Has Limits.

- In normal programming, we do this:
  - Rules + Data ⟶ Output
- But many real-world problems don't have clear rules.
- Example: How do you write exact rules to:
  - recognize a face?
  - understand handwriting?
  - decide if an email is spam?
- Writing rules for every possible case is impossible.
- ML solution:

  Data + Learning ⟶ Rules (model) ⟶ Output

2. Data Is Exploding.

- Today we generate huge amounts of data:
  - social media posts
  - sensor data
  - medical records
  - online transactions
- Humans cannot manually analyze this scale of data.
- ML helps computers automatically find patterns in large datasets.

Why is Machine Learning needed?

3. Systems Need to Improve Over Time.

- Traditional programs:
  - behave the same forever
  - don't improve unless reprogrammed
- Machine Learning systems:
  - learn from new data
  - improve accuracy over time
- Example:
  - A recommendation system gets better as you watch more content.

4. Many Problems Are Too Complex for Humans.

- Some problems are:
  - highly dynamic uncertain
  - influenced by many variables
- Examples:
  - weather prediction
  - stock market trends
  - disease diagnosis
- ML can handle complex relationships that humans can't easily model.

5. Automation at Scale: fast prediction; real time responses; automatic decision making

Differentiate ML from traditional programming

Factor	Traditional Programming	Machine Learning
Instruction Method	Explicit rules and logic	Learns patterns from data
Data Handling	Structured, predictable data	Large, often unstructured data
Outcome Predictability	Always the same result for same input	Predictions may vary as model adapts to new data
Flexibility	Limited to predefined conditions	Adjusts based on new data (self-improving)
Development Process	Linear: write, debug, deploy	Iterative: train, evaluate, tune, retrain
Transparency	Easy to trace and debug	Can be opaque, often needs explainable AI
Problem Complexity	Best for simple, well-defined tasks	Best for complex, data-rich tasks

Traditional programming is like following a recipe exactly, while Machine Learning is like learning to cook better by tasting food again and again.

Real-Life Examples of ML

- Spam Email Filtering – learns from past emails to block spam
- Face Unlock on Phones – recognizes faces even with glasses or beard
- Search Suggestions – predicts what you're typing
- YouTube / Netflix Recommendations – suggests content based on watch history
- Self-Driving Cars – detect lanes, vehicles, pedestrians
- Google Maps Traffic Prediction – predicts congestion
- Uber / Ola Fare Estimation – predicts trip cost & demand
- Disease Diagnosis – detects cancer from X-rays or MRI scans
- Disease Risk Prediction – predicts diabetes, heart disease
- Drug Discovery – speeds up molecule discovery
- Fraud Detection – detects unusual transactions
- Loan Approval Systems – evaluates credit risk
- Stock Market Analysis – trend prediction (not guarantees!)
- Amazon Product Recommendations
- Personalized Ads on social media
- Online Learning Platforms
- Voice Assistants (Alexa, Google Assistant)
- Handwriting Recognition

History of Machine Learning

- **1940s–1950s: Foundations of AI & Learning Ideas**
  - Early computers inspired scientists to think: "Can machines learn like humans?"
  - Mathematical and logical models of intelligence were proposed.
  - Learning was still a theoretical concept.
- **1959: Birth of Machine Learning (Term Coined)**
  - Arthur Samuel introduced the term Machine Learning.
  - He developed a checkers-playing program that improved by playing repeatedly.
- Milestone Quote (Concept): A machine that learns from experience.
- **1960s–1970s: Rule-Based Systems Era**
  - Focus on expert systems using:
    - IF–THEN rules
    - Human knowledge encoded manually
  - Worked well for small, controlled problems..But, failed for complex, real-world scenarios.

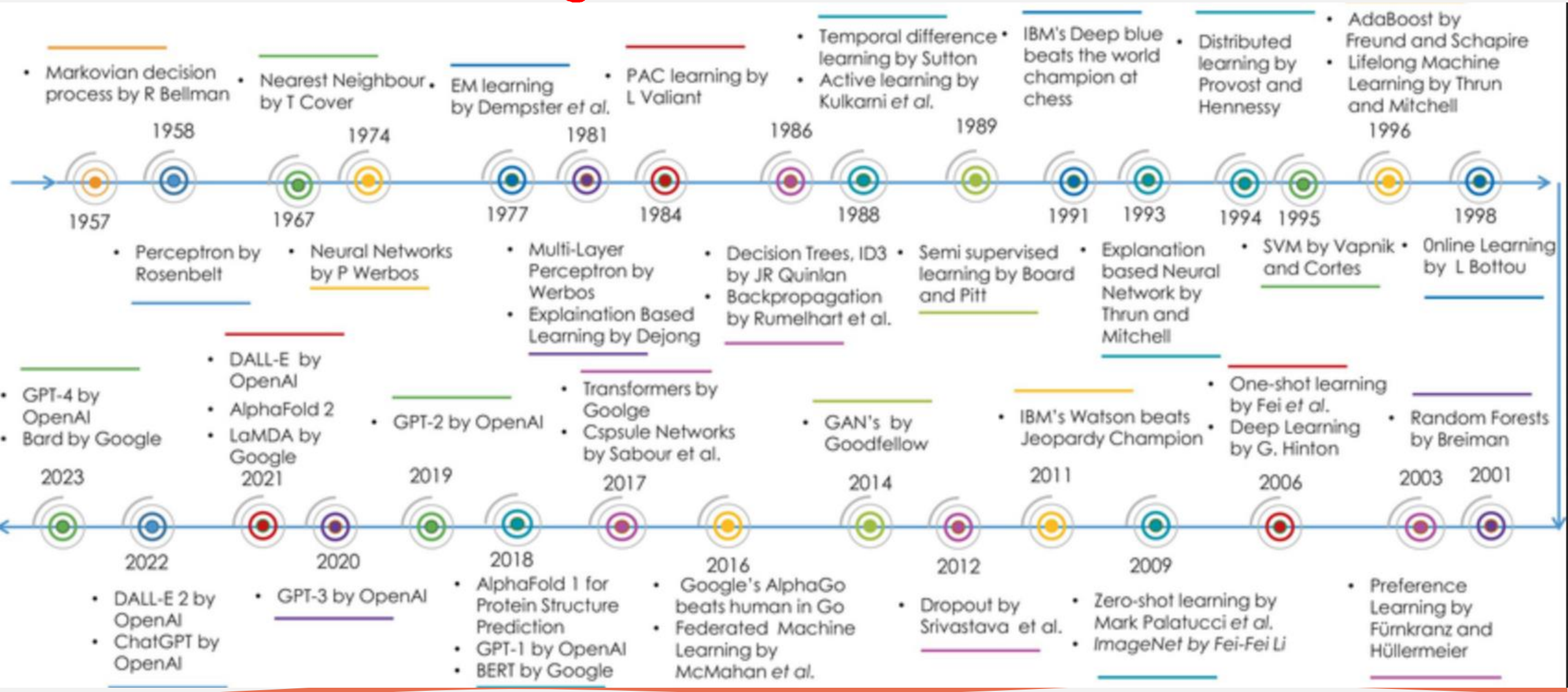
History of Machine Learning

- **1980s: Rise of Neural Networks**
  - Introduction of Artificial Neural Networks (ANNs) inspired by the human brain.
  - Backpropagation algorithm revived interest in learning systems.
  - Limited by low computing power and small datasets.
- **1990s: Statistical Learning Takes Over**
  - Shift from rules to data-driven methods.
  - Algorithms like: Decision Trees; k-Nearest Neighbors; Support Vector Machines
  - ML began to be used in speech and handwriting recognition.
- **2000s: Big Data + Better Computers**
  - Explosion of: Internet data; Storage capacity; Faster processors
  - ML became more practical and scalable.
  - Used in search engines and recommendation systems.

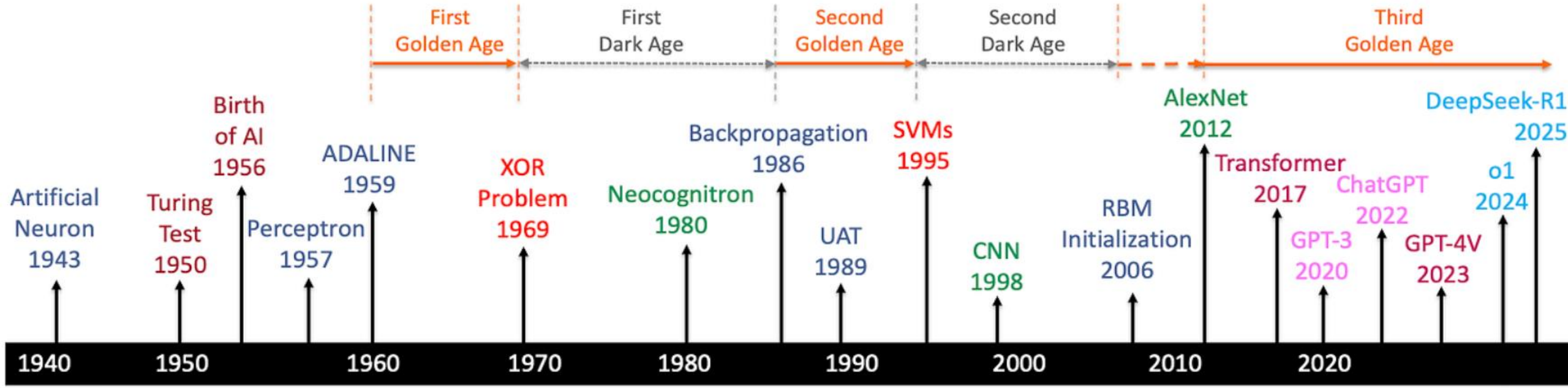
History of Machine Learning

- **2010s–Present: Deep Learning & Modern ML**
- Breakthroughs in Deep Learning using multi-layer neural networks.
- Powered by : GPUs; Massive datasets;
- Major successes in:
    - Image recognition
    - Speech recognition
    - Self-driving cars
    - Chatbots and assistants

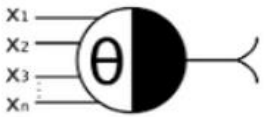
History of Machine Learning

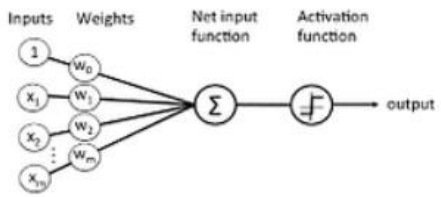


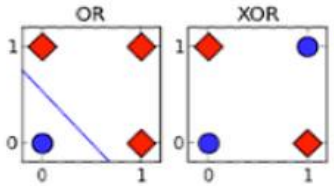
History of Machine Learning Techniques

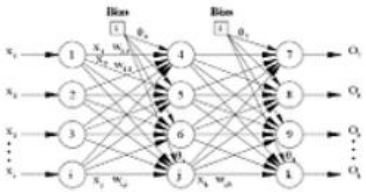


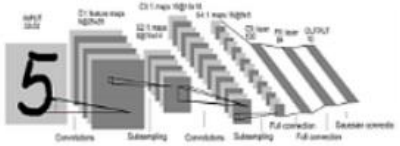
McCulloch-Pitts | Rosenblatt | Widrow-Hoff | Minsky-Papert | Rumelhart, Hinton et al. | LeCun | Hinton-Ruslan | Krizhevsky et al. | Vaswani

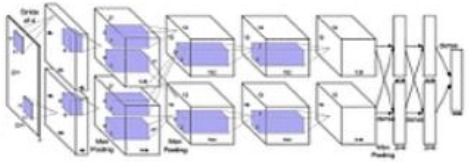












Key Factors Driving ML Growth

- **Massive Volumes of Data:** The digital age generates enormous amounts of data daily from web interactions, IoT devices, and various business operations. ML models require large, high-quality datasets to learn patterns and make accurate predictions, making this data availability a fundamental driver of growth.
- **Increased Computational Power:** Advancements in hardware, particularly the development of powerful Graphics Processing Units (GPUs) and specialized AI chips, have made it possible to process and analyze these vast datasets in a reasonable amount of time. Cloud computing platforms have further democratized access to this scalable infrastructure, reducing the need for heavy upfront investments.
- **Algorithm Improvements:** Continuous innovation in ML algorithms and techniques, such as deep learning and reinforcement learning, has significantly improved model accuracy and performance. These more sophisticated models can solve complex problems that were previously out of reach.
- **Automation and Efficiency Demands:** Businesses are increasingly adopting ML to automate repetitive and time-consuming tasks, leading to improved operational efficiency, cost reductions, and higher productivity.

Key Factors Driving ML Growth

- **Demand for Personalized Experiences:** ML enables companies to analyze user behavior and provide tailored recommendations and customized services in real time, which is a significant factor in customer satisfaction and retention for industries like e-commerce and media.
- **Wide Range of Applications and use Cases:** ML has proven valuable across diverse sectors, including healthcare (disease diagnosis), finance (fraud detection), and manufacturing (predictive maintenance), leading to widespread adoption as organizations recognize its potential to solve real-world problems and gain a competitive advantage.
- **Talent and Expertise Growth:** While skill gaps still exist, there is a growing global workforce and increased investment in training, making it easier for businesses to implement and manage ML solutions.

Advantages of Machine Learning

- Automation of Tasks
  - ML automates repetitive and complex tasks without continuous human intervention.
- Ability to Learn from Data
  - Systems improve performance as more data becomes available.
- Handling Complex Problems
  - ML can solve problems where rules are difficult or impossible to define.
- High Accuracy (with good data)
  - ML models often outperform traditional methods in prediction tasks.
- Scalability
  - Can process and analyze large volumes of data efficiently.
- Personalization
  - Powers customized user experiences in recommendations, marketing, and services.

Disadvantages of Machine Learning

- Requires Large Amount of Data
  - Insufficient data leads to poor performance.
- High Computational Cost
  - Training ML models can be time-consuming and resource-intensive.
- Lack of Interpretability
  - Many ML models work as “black boxes”.
- Bias in Results
  - Biased training data leads to biased predictions.
- Maintenance Overhead
  - Models need regular retraining as data patterns change.

Advantages of Machine Learning

- Automation of Repetitive Tasks
- Improved Decision-Making
- Identifies trends and enhances predictions.
- Continuous Improvement
- Handles large datasets efficiently.
- Reduces operational expenses.
- Utilizes technological advancements.

VS

Disadvantages of Machine Learning

- Requires high-quality data.
- High Computational Costs.
- Risk of Bias
- Replaces repetitive tasks.

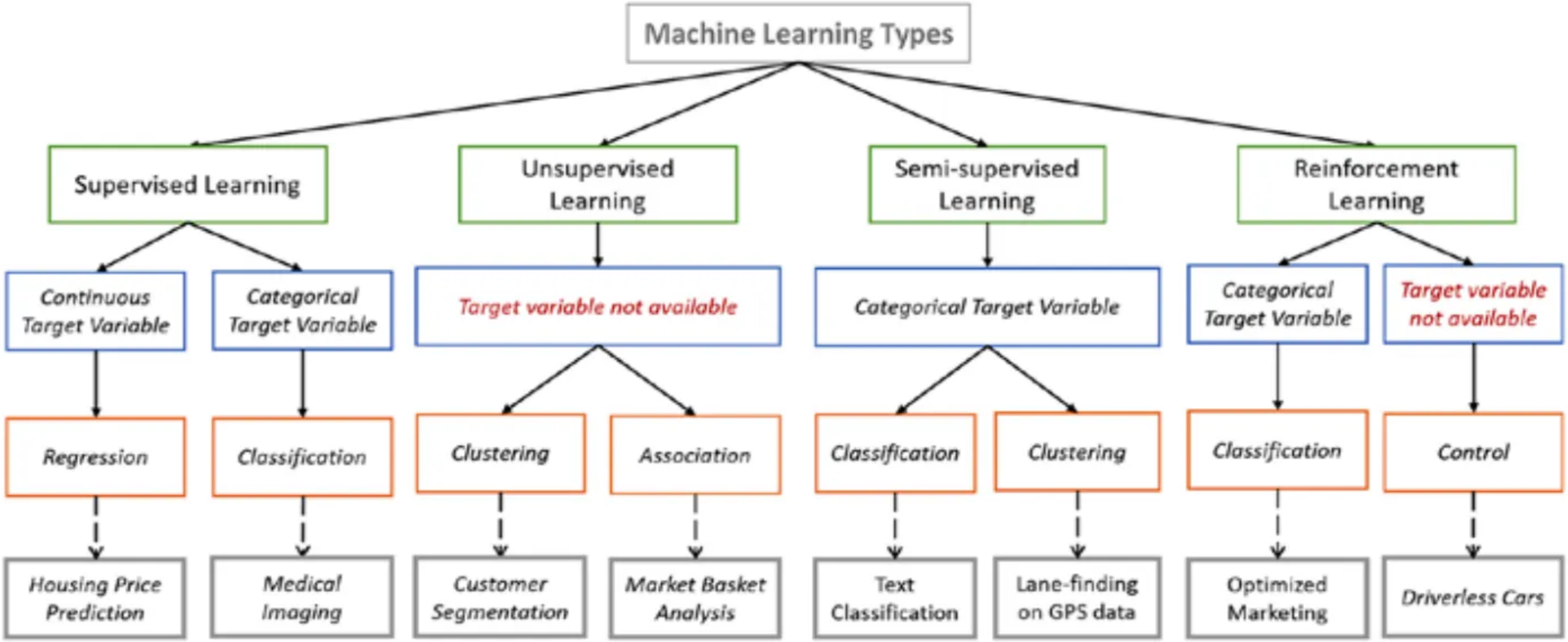
Limitations of Machine Learning

- Dependent on Data Quality
  - Poor-quality or noisy data reduces effectiveness.
- Limited Generalization
  - ML models struggle with situations outside their training data.
- No Common Sense or Reasoning
  - ML systems lack human intuition and understanding.
- Ethical and Privacy Concerns
  - Use of personal data can raise serious ethical issues.
- Not Suitable for All Problems
  - Simple rule-based tasks do not need ML.

Types of Machine Learning problems

- A machine learning problem is defined by:
  - Type of data available
  - Nature of output required
  - How learning happens
- Based on this, ML problems are broadly classified as follows:
  - Supervised Learning Problems $\longrightarrow$ Classification & Regression
  - Unsupervised Learning Problems $\longrightarrow$ Clustering & Association
  - Reinforcement Learning Problems $\longrightarrow$ Trial + Reward
  - Semi-Supervised Learning Problems

Types of Machine Learning problems



Types of Machine Learning problems

ML Problem Type	Data Type	Output Type	Example
Classification	Labeled	Categorical	Spam detection
Regression	Labeled	Numerical	Price prediction
Clustering	Unlabeled	Groups	Customer segments
Association	Unlabeled	Rules	Product bundling
Reinforcement	Feedback-based	Actions	Game playing

Types of Machine Learning

- Learning can be broadly classified into three categories, as mentioned below, based on the nature of the learning data and interaction between the learner and the environment.
- **Supervised Learning** – Supervised learning is a type of machine learning where an algorithm learns from labeled training data, and makes predictions or decisions based on that learning. In supervised learning, the algorithm is provided with input-output pairs, and the goal is to learn a mapping function from inputs to outputs. It is called 'supervised' because the process of an algorithm learning from the training dataset can be compared to a teacher supervising the learning process.
- Example: Image recognition, language translation.

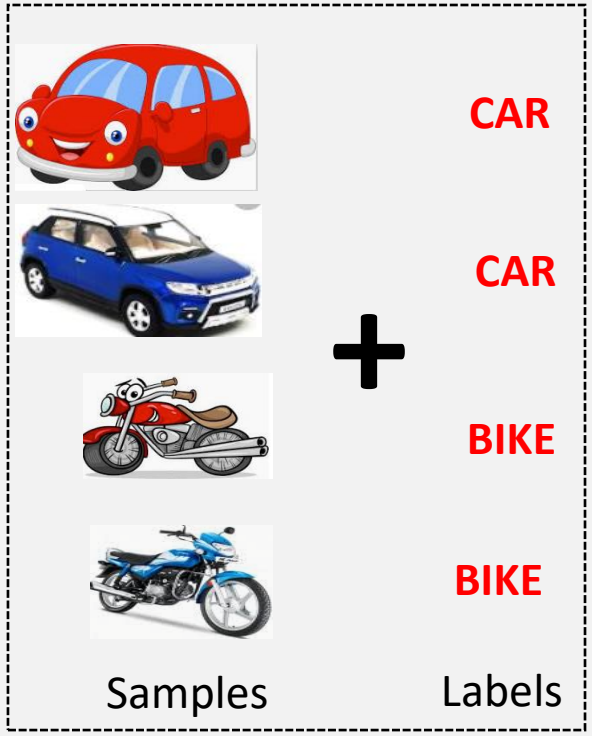
- **Unsupervised Learning** – Unsupervised learning involves training algorithms with unlabeled data. The system tries to learn the patterns and the structure from the data without any supervision. The algorithms work to explore the data and can automatically find patterns or relationships within it.
- Example: Clustering for customer segmentation, anomaly detection.

Types of Machine Learning

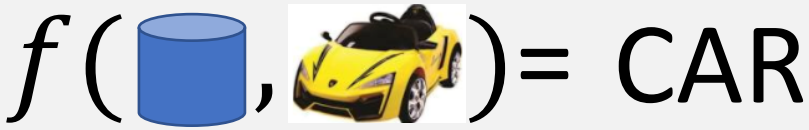
- **Semi-supervised Learning** – Semi-supervised learning is a hybrid approach where the algorithm is trained on a dataset that contains both labeled and unlabeled data. It's useful when obtaining labeled data is expensive or time-consuming. The algorithm learns from the small amount of labeled data and the larger amount of unlabeled data to make predictions.
- Example: Sentiment analysis, speech recognition.

- **Reinforcement Learning** – Reinforcement learning is a type of machine learning where an agent learns how to behave in an environment by performing certain actions and receiving rewards or penalties in return. The agent learns to achieve a goal in an uncertain, potentially complex environment. Reinforcement learning is based on the idea of agents taking actions in an environment to maximize some notion of cumulative reward.
- Example: Game playing (e.g., AlphaGo), robotic control.

What is Supervised Learning?



= **Training Dataset**



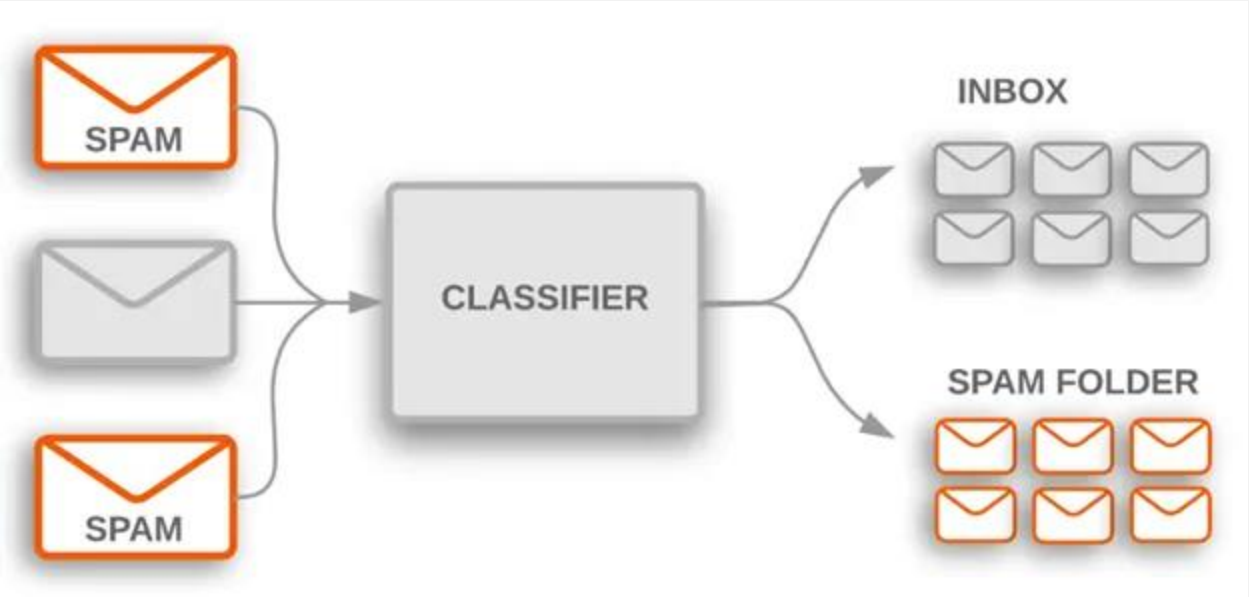
- Supervised Learning is a type of Machine Learning in which a model is trained using labeled data, where each input is paired with a known correct output, and the model learns to map inputs to outputs.
- Supervised learning is a machine learning technique where models are trained on labeled datasets to predict correct outputs for new inputs.

Types of Supervised Learning

- The two primary types of supervised learning are **classification and regression**. These categories define the nature of the output the model is designed to predict.
- **Classification :** Classification algorithms are used when the output variable is a **categorical value or discrete label**. The goal is to assign an input data point to a specific, predefined class.
- Examples:
  - **Spam detection:** Classifying an email as either "spam" or "not spam" (binary classification).
  - **Image recognition:** Identifying whether an image contains a "cat" or "dog".
  - **Medical diagnosis:** Predicting if a tumor is "malignant" or "benign".
- **Regression :** Regression algorithms are used when the output variable is a **continuous numerical value**. The goal is to predict a quantity or a point within a range.
- Examples:
  - **Real estate:** Predicting the price of a house based on its size, location, and age.
  - **Forecasting:** Predicting future stock prices or weather temperatures.
  - **Risk assessment:** Estimating the likelihood of a customer defaulting on a loan.

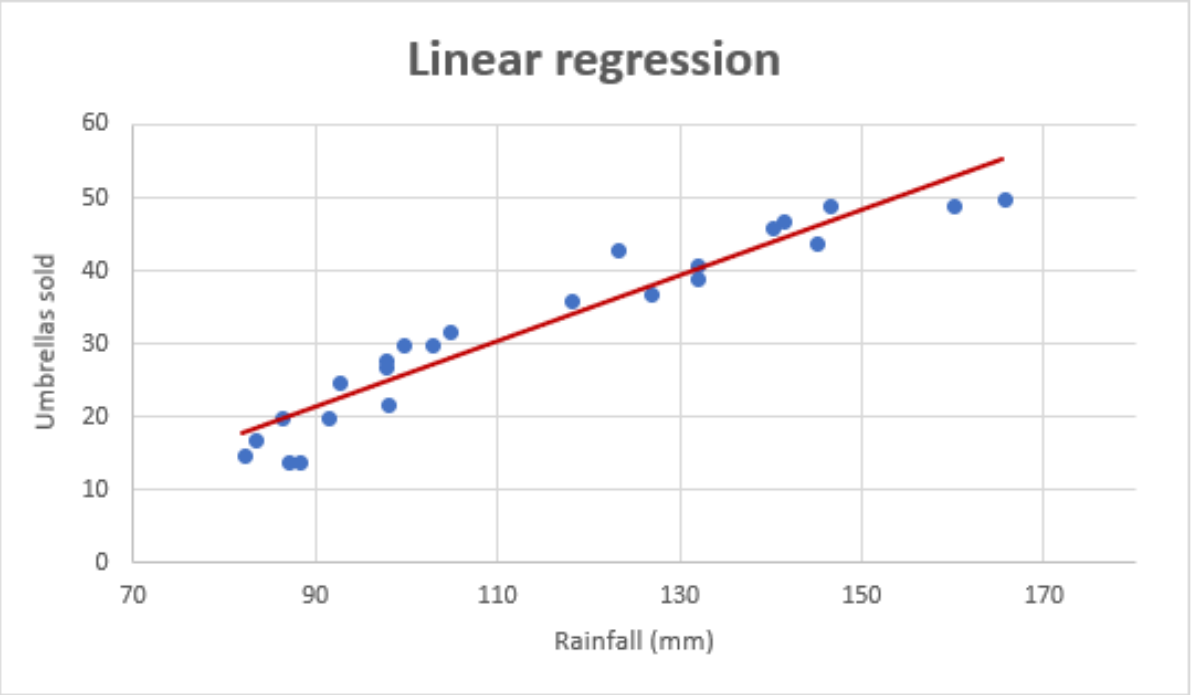
Classification

- Classification is a process of categorizing a given set of data into classes, It can be performed on both structured or unstructured data. The process starts with predicting the class of given data points. The classes are often referred to as target, label or categories.
- In machine learning and statistics, classification is the problem of identifying to which of a set of categories (sub-populations) a new observation belongs, on the basis of a training set of data containing observations (or instances) whose category membership is known.

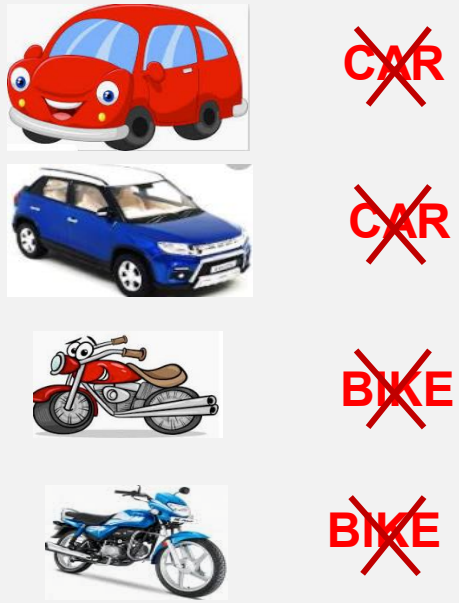


Regression

- Regression in machine learning is a supervised learning technique used to predict continuous numerical values by learning relationships between input variables (features) and an output variable (target).
- The main goal of regression analysis is to plot a line or curve that best fit the data and to estimate how one variable affects another.
- It helps understand how changes in one or more factors influence a measurable outcome and is widely used in forecasting, risk analysis, decision-making and trend estimation.



What is Un-supervised Learning?



Dataset

Clustering

- Unsupervised Learning is a type of Machine Learning in which a model is trained on unlabeled data, and the system learns to discover hidden patterns, structures, or relationships in the data without any predefined output.
- The machine learns without a teacher, by finding patterns on its own.

Clustering

- Clustering is an unsupervised learning technique used to group similar data points together based on their features, without using any labeled output.
- Clustering automatically forms groups where items inside a group are more similar to each other than to items in other groups. Popular algorithm used are the K-means clustering & Hierarchical Clustering.
- Worked As:
  - Only input data is given (no labels)
  - The algorithm measures similarity (distance, patterns)
  - Similar data points are grouped together
  - Each group is called a cluster
- Used For:
  - To discover hidden patterns
  - To organize large datasets
  - To perform data exploration
  - To support decision-making

Association Rule Learning

- Association Rule Learning is an unsupervised learning technique used to discover interesting relationships, patterns, or associations among items in large datasets.
- It finds out what items frequently occur together.
- Popular algorithm used is Apriori Algo.
- Works As:
  - A dataset of transactions is given.
  - The algorithm finds frequently occurring item combinations.
  - Rules are generated in the form: IF X occurs ⟶ THEN Y occurs.
- Used For:
  - Market basket analysis
  - Product recommendation
  - Website click analysis
  - Medical diagnosis patterns

Supervised Vs Un-supervised Learning

Basis	Supervised Learning	Unsupervised Learning
Definition	Supervised learning algorithms train data, where every input has a corresponding output.	Unsupervised learning algorithms find patterns in data that has no predefined labels.
Goal	The goal of supervised learning is to predict or classify based on input features.	The goal of unsupervised learning is to discover hidden patterns, structures and relationships.
Input Data	Labeled: Input data with corresponding output labels.	Unlabeled: Input data is raw and unlabeled.
Human Supervision	Supervised learning algorithms needs human supervision to train the model.	Unsupervised learning algorithms does not any kind of supervision to train the model..
Tasks	Regression, Classification	Clustering, Association and Dimensionality Reduction
Complexity	supervised machine learning methods are computationally simple.	Unsupervised machine learning methods are computationally complex.
Algorithms	Linear regression, K-Nearest Neighbors, Decision Trees, Naive Bayes, SVM	K- Means clustering, DBSCAN, Autoencoders
Accuracy	Supervised machine learning methods are highly accurate.	Unsupervised machine learning methods are less accurate.
Applications	Image classification, Sentiment Analysis, Recommendation systems	Customer Segmentation, Anomaly Detection, Recommendation Engines, NLP

Classification Vs Clustering

Classification	Clustering
Uses labelled data as the input	Uses unlabelled data as the input
The output is known	The output is unknown
Uses supervised machine learning	Uses unsupervised machine learning
A training data set is provided and used to produce classifications	A training data set is not provided and used to produce clusters
Examples of algorithms: Decision-trees, Bayesian Classifiers and Support Vector Machines (SVM)	Examples of algorithms: Partition-based clustering (k-means), Hierarchical clustering (agglomerative & divisive) and DBSCAN
Can be more compex than clustering	Can be less compex than classification
Does not specify areas for improvement	Specifies areas for improvement
Two-phase	Single-phase
Boundary conditions must be specified	Boundary conditions do not always need to be specified

Classification Vs Regression

Regression Algorithm	Classification Algorithm
The output variable has to be a real value or continuous in nature.	The output variable is discrete in the Classification SML problem.
Regression algorithm helps to map the input value and the continuous output variable.	Classification algorithm helps in mapping the input value with the output variable which is discrete in nature.
Regression algorithms are only used for data that is continuous.	Classification algorithms are only used for data that is discrete.
Finds the best fit for accurately predicting the output data in Regression.	Finds the decision boundary which helps in classifying the dataset into different classes in Classification
Regression algorithms help in solving the problems like predicting the weather or the price of Houses in a neighbourhood according to a real estate trend.	Classification algorithms are useful for problems like identifying spam emails, recognition of speech, identifying cancer cells, etc.

What is Reinforcement Learning?

- Reinforcement Learning is a type of Machine Learning where an agent learns by interacting with an environment, taking actions, and receiving rewards or penalties based on those actions.
- The machine learns by trial and error, just like a child learning to ride a bicycle.
- Works As:
  - Agent performs an action
  - Action affects the Environment
  - Environment gives a Reward or Penalty
  - Agent learns which actions give maximum reward
  - Process repeats until optimal behavior is learned
- Used For:
  - Game playing (chess, video games)
  - Robotics (path planning)
  - Self-driving cars
  - Traffic signal control
  - Recommendation systems (adaptive feedback)

What is Reinforcement Learning?

- **Core Concepts:**
  - Agent: The decision-maker (e.g., a robot, game AI).
  - Environment: The world the agent interacts with (e.g., a maze, game board).
  - State: A snapshot of the environment at a given time.
  - Action: A choice the agent makes.
  - Reward: A numerical signal indicating the quality of an action (positive for good, negative for bad).
  - Policy: The agent's strategy for choosing actions in different states.